

COMPREHENSIVE AUTOMATED TEST EXAM 2026

Examination Year: 2026 | Maximum Marks: **100** | Total Questions: **2** | Time Allowed: **3 Hours**

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	Total
Marks	50	50	100

Question 1

[50 Marks]

Calculate the magnetic flux $\Phi = \int \mathbf{B} \cdot d\mathbf{A}$ through loop with $r=0.5\text{m}$ in field $B=2.0\text{T}$.

Electric Field Lines (+q)

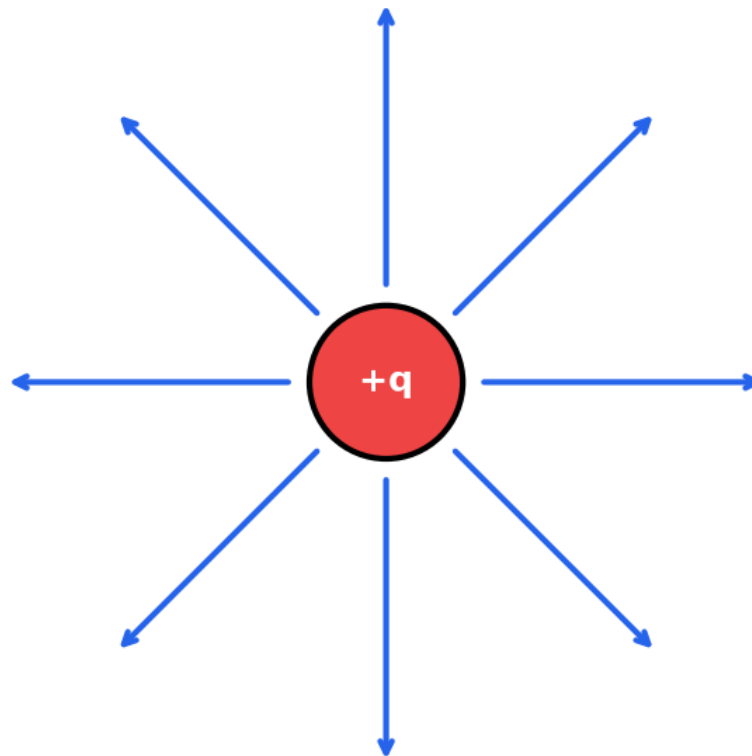


Figure 1

Question 2

[50 Marks]

Explain prokaryotic DNA replication mechanism.
